

Accuracy of Multiply-Upstream, Semi-Lagrangian Advective Schemes

A. McDONALD

Irish Meteorological Service, Dublin

28 May 1983 and 9 March 1984

ABSTRACT

The stability and accuracy of the multiply-upstream, semi-Lagrangian method of integrating the advective equation in two dimensions is examined for four different interpolation schemes; namely, bilinear, biquadratic, bicubic and biquartic. All are shown to be consistent and unconditionally stable for constant advecting velocity. Their respective amplitude and phase errors are discussed. They are then used to integrate the test problem of a cone being advected about the plane at constant angular velocity. The merits of the schemes relative to each other and relative to a well tried Eulerian scheme are examined with particular regard to accuracy and computation time.

1. Introduction

It has recently been shown that the semi-Lagrangian approach offers advantages in integrating the advection part of the primitive meteorological equations. In particular, going multiply upstream offers the possibility of unconditionally stable integrations.

With this approach, Robert (1981, 1982) obtained stable integrations of both the divergent and nondivergent shallow water equations with a timestep of up to 2 h, using a semi-Lagrangian, semi-implicit model. This timestep is four to six times as large as the maximum permitted timestep for an equivalent Eulerian semi-implicit model. The resultant potential improvement in efficiency was partially offset by the expense of the bicubic and biquartic interpolation schemes used to integrate along the trajectory.

Bates and McDonald (1982), having shown why it gives unconditionally stable integrations, used the same approach to integrate a multilevel primitive equation model. They used a split explicit semi-Lagrangian scheme, and obtained stable and accurate integrations with a timestep four times as large as that allowed by an equivalent split explicit Eulerian scheme, which resulted in a large decrease in computation time. They showed that a biquadratic interpolation scheme gave forecasts as accurate as the Eulerian scheme, and also that the integrations were stable for a timestep eight times as large as that of the Eulerian, although the forecasts were less accurate with this larger timestep.

This note seeks to clarify and to elaborate upon some of the issues raised by the aforementioned papers, and in particular to compare the different interpolation schemes to each other and to some well-tested Eulerian scheme in order to evaluate their relative merits with regard to stability, accuracy and efficiency. In Section 2 a stability analysis of the various multiply-upstream,

semi-Lagrangian schemes is carried out for a constant flow. In Section 3, a numerical integration is performed for a problem whose solution is known analytically, namely, the advection of a scalar in a fluid rotating at constant angular velocity, using the various schemes.

2. Stability analysis

a. The multiply-upstream method

The equation describing the advection of a scalar is

$$\partial\psi/\partial t + \mathbf{v} \cdot \nabla\psi = 0, \quad (1)$$

where $\psi = \psi(\mathbf{r}, t)$ is a scalar function of position and time and $\mathbf{v} = \mathbf{v}(\mathbf{r}, t)$ is the advection velocity. If $\mathbf{v}$ is a constant, then the solution of Eq. (1) at time t is known in terms of the initial conditions:

$$\psi(\mathbf{r}, t) = \psi(\mathbf{r} - \mathbf{v}t, 0) = \psi(\mathbf{r}_*, 0), \quad (2)$$

where $\mathbf{r}_* = \mathbf{r} - \mathbf{v}t$ is the departure point. The multiply-upstream semi-Lagrangian method of solving Eq. (1) by finite differences rests on the assumption that Eq. (2) still holds to a very good approximation for non-constant velocity.

In a two-dimensional rectangular coordinate system, we proceed as follows. Writing $x = I\Delta x$, $y = J\Delta y$ and $t = n\Delta t$ and $\mathbf{v} = (u, v)$, then, using Eq. (2) we obtain

$$\psi\{I\Delta x, J\Delta y, (n+1)\Delta t\} = \psi(x_*, y_*, n\Delta t), \quad (3)$$

where

$$x_* = I\Delta x - u\Delta t; \quad y_* = J\Delta y - v\Delta t. \quad (4)$$

Introducing the notation $\psi_{I,J}^n = \psi(I\Delta x, J\Delta y, n\Delta t)$, the function $\psi(x_*, y_*, t)$ is approximated by a Lagrange interpolating polynomial (Carnahan *et al.*, 1969; p.

65) using values of ψ at the points nearest to x_* and y_* :

$$\psi(x_*, y_*, t) = \sum_{\mu} \sum_{\nu} W_{\mu\nu} \psi_{\mu\nu}^n,$$

$$W_{rs} = \prod_{\mu \neq r} \frac{(x_* - x_{\mu})}{(x_r - x_{\mu})} \prod_{\nu \neq s} \frac{(y_* - y_{\nu})}{(y_s - y_{\nu})}. \quad (5)$$

The subscripts μ and ν range over the points being used in the interpolation as

$$\left. \begin{array}{ll} \text{Bilinear;} & \mu: i-1, i; \quad \nu: j-1, j \\ \text{Biquadratic;} & \mu: i-1, i, i+1; \quad \nu: j-1, j, j+1 \\ \text{Bicubic;} & \mu: i-2, i-1, i, i+1; \quad \nu: j-2, j-1, j, j+1 \\ \text{Biquartic;} & \mu: i-2, i-1, i, i+1, i+2; \quad \nu: j-2, j-1, j, j+1, j+2 \end{array} \right\}. \quad (6)$$

The points (i, j) are chosen for the bilinear and bicubic schemes such that

$$\begin{aligned} (i-1)\Delta x < x_* \leq i\Delta x, \\ (j-1)\Delta y < y_* \leq j\Delta y, \end{aligned} \quad (7)$$

and for the biquadratic and biquartic schemes such that

$$\begin{aligned} \left(i - \frac{1}{2}\right)\Delta x < x_* \leq \left(i + \frac{1}{2}\right)\Delta x, \\ \left(j - \frac{1}{2}\right)\Delta y < y_* \leq \left(j + \frac{1}{2}\right)\Delta y. \end{aligned} \quad (8)$$

If $\alpha, \hat{\alpha}, \beta, \hat{\beta}$ are defined as

$$\left. \begin{aligned} \alpha &= u\Delta t/\Delta x, & \beta &= v\Delta t/\Delta y \\ \hat{\alpha} &= \alpha + i - I, & \hat{\beta} &= \beta + j - J \end{aligned} \right\}, \quad (9)$$

then the choices of i and j given in Eq. (7) and (8) guarantee that, for the bilinear and bicubic cases,

$$0 \leq \hat{\alpha} < 1 \quad \text{and} \quad 0 \leq \hat{\beta} < 1 \quad (10)$$

and that, for the biquadratic and biquartic cases,

$$-0.5 \leq \hat{\alpha} < 0.5 \quad \text{and} \quad -0.5 \leq \hat{\beta} < 0.5. \quad (11)$$

It will be shown in Section 2c that these conditions are sufficient to ensure unconditional stability for all four interpolation schemes.

b. Consistency

It is important that a finite difference approximation to a differential equation be consistent, since this property allied with that of stability implies convergence via the Lax Equivalence Theorem (Richtmeyer and Morton, 1967). Consider the one-dimensional version of Eq. (1), that is,

$$d\psi/dt = \partial\psi/\partial t + u\partial\psi/\partial x = 0. \quad (12)$$

The question to be answered is whether the following finite difference equation,

$$\Delta\psi/\Delta t = \{\psi[I\Delta x, (n+1)\Delta t] - \psi(x_*, n\Delta t)\}/\Delta t = 0 \quad (13)$$

approaches Eq. (12) in the limit as Δt approaches zero.

Using the Lagrange interpolating polynomial to approximate $\psi(x_*, n\Delta t)$ and subsequently writing a Tay-

lor series expansion for the various terms generated, we get the following result. (The subscript r takes values corresponding to the interpolation scheme used as follows, linear: 1; quadratic: 2; cubic: 3; and quartic: 4.)

$$\begin{aligned} \frac{\Delta\psi}{\Delta t} &= D_r\psi^n + \sum_{l=r+1}^{\infty} \frac{1}{l!} \\ &\times \left[\Delta t^{l-1} \left\{ \frac{\partial^l \psi}{\partial t^l} \right\}_I - \frac{\Delta x^l}{\Delta t} \left\{ \frac{\partial^l \psi}{\partial x^l} \right\}_I W_r'(\hat{\alpha}, p) \right], \end{aligned} \quad (14)$$

where $p = \alpha - \hat{\alpha}$, and $D_r\psi$ and $W_r'(\hat{\alpha}, p)$ are listed in the Appendix.

In taking the limit $\Delta t \rightarrow 0$ there are three cases to consider.

1) Case 1. The grid is refined in such a way that $\Delta t/\Delta x \rightarrow 0$ as $\Delta t \rightarrow 0$. In this limit $p = 0$ and $\hat{\alpha} = \alpha$; that is, the departure point x_* , approaches the arrival point $I\Delta x$. Because

$$\begin{aligned} W_r'(\alpha, 0) &= -u\Delta t/\Delta x \{1 + O[\Delta t/\Delta x] \\ &+ \dots + O[(\Delta t/\Delta x)^{r-1}]\}, \end{aligned} \quad (15)$$

it is obvious from Eq. (14) that all four schemes are consistent in this limit.

2) Case 2. As $\Delta t \rightarrow 0$ the grid is refined in such a way that $\Delta t/\Delta x = \gamma$, a constant. In this limit, $\alpha \rightarrow u\gamma$ and $\hat{\alpha} \rightarrow u\gamma - p$; i.e., the distance between the departure point and the arrival point is held constant. So

$$\Delta x^l W_r'(\hat{\alpha}, p)/\Delta t \rightarrow \Delta x^{l-1} W_r'(u\gamma - p, p)/\gamma, \quad (16)$$

and all the schemes are consistent.

3) Case 3. As $\Delta t \rightarrow 0$ the grid is refined such that $\Delta x/\Delta t \rightarrow 0$, so that $\hat{\alpha}$ and p tend to infinity and the distance between the departure and the arrival points gets larger and larger. In this situation,

$$\begin{aligned} \Delta x^l W_r'(\hat{\alpha}, p)/\Delta t &\rightarrow \Delta t^{l-1} \{1 + O[\Delta x/\Delta t] \\ &+ \dots + O[(\Delta x/\Delta t)^{l-1}]\}. \end{aligned} \quad (17)$$

Again, all four schemes are consistent.

The truncation error of the r th scheme is $O(\Delta t^r) + O(\Delta x^r) + O(\Delta t\Delta x^a) + O(\Delta t^2\Delta x^b)$, where $a = 0$ for the linear, $a = 2$ for the quadratic and cubic, and

$a = 4$ for the quartic schemes, respectively. For the cubic, $b = 1$ and for the quartic, $b = 2$, and b is zero otherwise.

c. Phase and amplitude accuracy

In order to examine the stability of the various interpolation schemes for constant v the following solution to Eq. 1 is assumed for the two-dimensional rectangular case,

$$\psi_{IJ} = \psi_0 \lambda^n \exp\{i(kI\Delta x + lJ\Delta y)\} \quad (18)$$

and substituted into Eq. (3) using Eq. (5) to evaluate $\psi(x_*, y_*, t)$. In all four cases, λ splits into the product of an "x amplification factor" by a "y amplification factor" as follows,

$$\lambda = \hat{\Lambda}(\hat{\alpha}, k) \hat{\Lambda}(\hat{\beta}, l), \quad (19)$$

making the analysis particularly simple. A scheme is stable if both

$$|\hat{\Lambda}(\hat{\alpha}, k)| \leq 1 \quad \text{and} \quad |\hat{\Lambda}(\hat{\beta}, l)| \leq 1,$$

since this guarantees $|\lambda| \leq 1$.

$|\hat{\Lambda}(\hat{\alpha}, k)|^2$ for the linear ($\hat{\Lambda}_1$), quadratic ($\hat{\Lambda}_2$), cubic ($\hat{\Lambda}_3$), and quartic ($\hat{\Lambda}_4$) schemes is as follows, where

$$c = 1 - \cos k\Delta x, \quad (20)$$

$$|\hat{\Lambda}_1(\hat{\alpha}, k)|^2 = 1 - 2\hat{\alpha}(1 - \hat{\alpha})c, \quad (21)$$

$$|\hat{\Lambda}_2(\hat{\alpha}, k)|^2 = 1 - \hat{\alpha}^2(1 - \hat{\alpha}^2)c^2, \quad (22)$$

$$|\hat{\Lambda}_3(\hat{\alpha}, k)|^2 = 1 - \hat{\alpha}(2 - \hat{\alpha})(1 - \hat{\alpha}^2)c^2\{3 + 2c\hat{\alpha}(1 - \hat{\alpha})\}/9, \quad (23)$$

$$|\hat{\Lambda}_4(\hat{\alpha}, k)|^2 = 1 - \hat{\alpha}^2(1 - \hat{\alpha}^2)(4 - \hat{\alpha}^2)c^3\{4 + c(1 - \hat{\alpha}^2)\}/36. \quad (24)$$

As can be seen, $|\hat{\Lambda}_1(\hat{\alpha}, k)|^2 \leq 1$ and $|\hat{\Lambda}_3(\hat{\alpha}, k)|^2 \leq 1$ as long as $0 \leq \hat{\alpha} < 1$; and $|\hat{\Lambda}_2(\hat{\alpha}, k)|^2 \leq 1$ and $|\hat{\Lambda}_4(\hat{\alpha}, k)|^2 \leq 1$ provided $0 \leq \hat{\alpha}^2 < 1$. Similarly, $|\hat{\Lambda}_1(\hat{\beta}, l)|^2 \leq 1$ and $|\hat{\Lambda}_3(\hat{\beta}, l)|^2 \leq 1$ for $0 \leq \hat{\beta} < 1$; and $|\hat{\Lambda}_2(\hat{\beta}, l)|^2 \leq 1$ and $|\hat{\Lambda}_4(\hat{\beta}, l)|^2 \leq 1$ for $0 \leq \hat{\beta}^2 < 1$ as can be seen by replacing $\hat{\alpha}$ by $\hat{\beta}$ and k by l in Eqs. (20)–(24). Hence, all four schemes are unconditionally stable since the conditions on $\hat{\alpha}$ and $\hat{\beta}$ are guaranteed by the choice of interpolation points given in Eq. (6).

The quantities $\hat{\alpha}$ and $\hat{\beta}$ defined in Eq. (9) can also be expressed for the bilinear and bicubic schemes as

$$\hat{\alpha} = \alpha - [\alpha] \quad \text{if} \quad \alpha \geq 0, \quad (25)$$

$$\hat{\alpha} = \alpha - [\alpha - 1] \quad \text{if} \quad \alpha < 0, \quad (26)$$

and for the biquadratic and biquartic schemes as

$$\hat{\alpha} = \alpha - [\alpha + 0.5] \quad \text{if} \quad \alpha \geq -0.5, \quad (27)$$

$$\hat{\alpha} = \alpha - [\alpha - 0.5] \quad \text{if} \quad \alpha < -0.5, \quad (28)$$

where $[x]$ is the integer part of x ; $\hat{\beta}$ is defined in exactly the same way, with β replacing α in all the above formulas. Substituting these values of $\hat{\alpha}$ in Eqs. (21)–

(24) gives a set of functions $|\Lambda_r(\alpha, k)| = |\hat{\Lambda}_r(\hat{\alpha}, k)|$, where $r = 1, 2, 3$ or 4 , having the following property

$$|\Lambda_r(\alpha + n, k)| = |\Lambda_r(\alpha, k)|, \quad (29)$$

where n is any integer.

The lines of constant $|\Lambda_r(\alpha, k)|$ are displayed in Fig. 1 for $0 \leq k\Delta x \leq \pi$ and $0 \leq \alpha \leq 1$. The plots for all other regions $n < \alpha \leq n + 1$ are identical to these plots because of Eq. (29).

All four schemes are most severely damped when $\alpha = 0.5$ and undamped when $\alpha = 0.0$ or 1.0 . The damping is heaviest for the shortest resolvable wavelengths and it decreases as the wavelength increases. The linear scheme is the most damped followed by the cubic, quadratic, and quartic. Notice that the strictures imposed on α by Eq. (8) mean that the quadratic and quartic cases have amplitudes which are symmetric about $\alpha = 0.5$. One consequence is a better amplitude representation than if, for instance, i had been chosen such that $(i - 1)\Delta x \leq x_* \leq i\Delta x$. Another is that the quadratic amplitude is less damped than the cubic for all wavelengths and all α .

The phase speed normalized to u is given by the following for the one dimensional case for all interpolation schemes.

$$R_r(\alpha, k) = p/\alpha + \phi_r/\alpha k\Delta x, \quad (30)$$

where the values of ϕ_r are given by

$$\tan \phi_1 = \hat{\alpha} \sin k\Delta x / (1 - \hat{\alpha}c), \quad (31)$$

$$\tan \phi_2 = \hat{\alpha} \sin k\Delta x / (1 - \hat{\alpha}^2c), \quad (32)$$

$$\tan \phi_3 = \hat{\alpha} \sin k\Delta x \{1 + c(1 - \hat{\alpha}^2)/3\} / \{1 - c\hat{\alpha}^2 - c^2\hat{\alpha}(1 - \hat{\alpha}^2)/3\}, \quad (33)$$

$$\tan \phi_4 = \hat{\alpha} \sin k\Delta x \{1 + c(1 - \hat{\alpha}^2)/3\} / \{1 - c\hat{\alpha}^2 - c^2\hat{\alpha}^2(1 - \hat{\alpha}^2)/6\}, \quad (34)$$

where, again, $\hat{\alpha}$ is given in Eqs. (25)–(28).

When $i \neq I$, $R_r(\alpha, k)$ tends to 1 as α tends to any nonzero integer. Also $R_r(\alpha, k) = R_r(-\alpha, k)$. In Figs. 2 the isolines of $R_r(\alpha, k)$ are plotted for $0 < \alpha \leq 2$ and $0 \leq k\Delta x < \pi$. Overall, the cubic interpolation scheme has the best phase representation followed in order by the quartic, the linear and the quadratic.

These results may be summarized for all schemes as,

- 1) the amplitude and phase errors both decrease as the wavelength increases, for a given α ,
- 2) averaged over each of the regions $0 \leq \hat{\alpha} < 1$, the phase error decreases as α increases, whereas the amplitude error remains unchanged,
- 3) close to $\alpha = 0$, the phase error is largest and the amplitude error is least, whereas close to all integer values of α the phase and amplitude errors are zero,
- 4) cubic interpolation gives the best phase representation and quartic interpolation gives the most faithful amplitude representation.

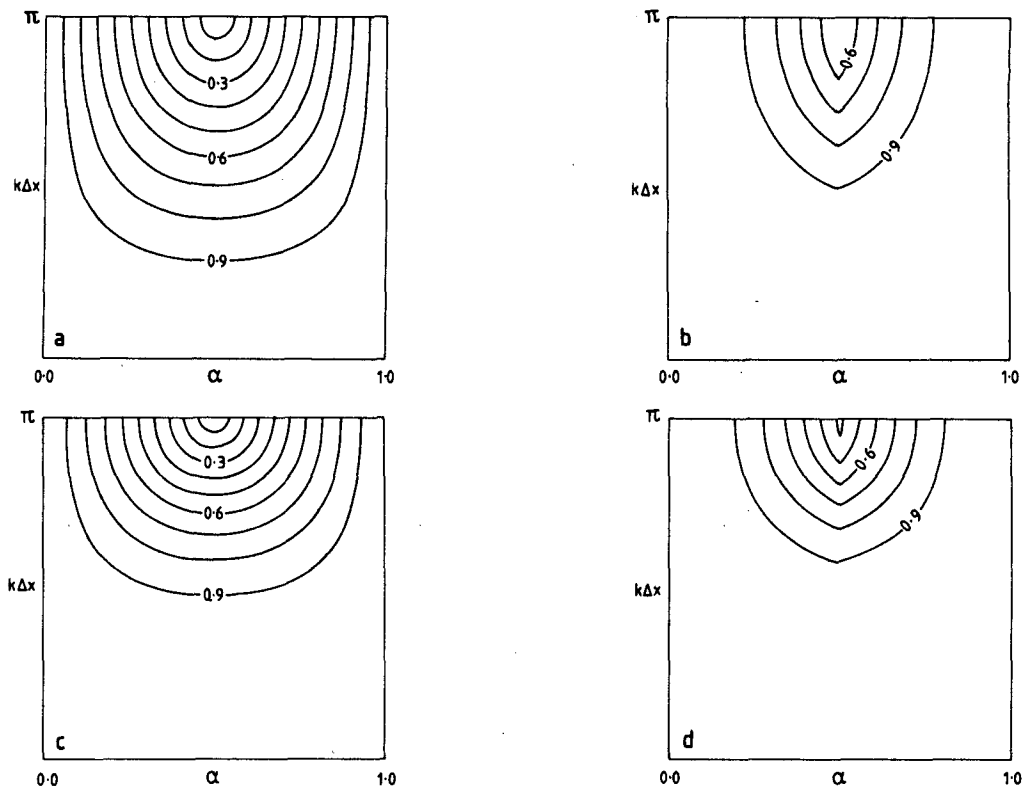


FIG. 1. Isolines of the amplification factor plotted as a function of $k\Delta x$ and α for (a) the linear, (b) the quadratic, (c) the cubic and (d) the quartic scheme.

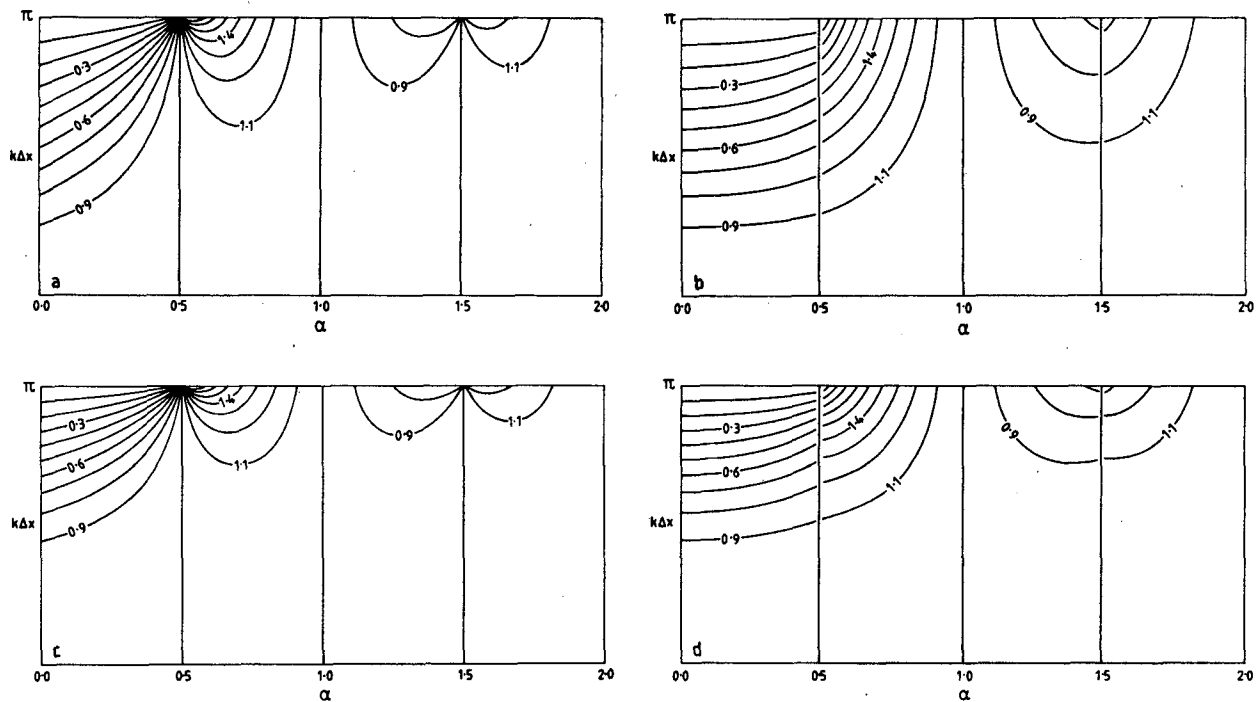


FIG. 2. Normalized phase speed isolines plotted as a function of $k\Delta x$ and α for (a) the linear, (b) the quadratic, (c) the cubic and (d) the quartic scheme.

From 2) and 3), the finite difference approximation to Eq. (1) is closer to the analytical solution for $\alpha \gg 1$ than for $\alpha \approx 0$, for constant velocity, if the multiply-upstream approach is taken.

3. Two-dimensional numerical tests

In this section, the accuracy of the various semi-Lagrangian schemes is examined and compared to a standard scheme; i.e., the Arakawa-Adams-Bashforth scheme which is described in detail in Orszag (1971) Section 2i. The test problem used for this purpose is that of advecting a cone at a constant angular velocity about a point in the (x, y) plane using Eq. (1) [see Crowley (1968) and Molenkamp (1968)].

Two complications may develop near the boundary of the area of integration. The upstream point (x_*, y_*) may be such that the values of (i, j) which satisfy Eqs. 7 or 8 are outside the range for which fields are defined. Even if they are inside, some of the points needed to perform the interpolation may be outside. The solution to the former problem which has been adopted here is to have a number of extra passive boundary lines at which the fields are defined initially, but never updated during the integration; this number is chosen such that it is the smallest integer greater than the maximum possible value of $|\alpha|$ or $|\beta|$. Having done this, the latter problem is solved by performing a bilinear interpolation whenever one of the higher interpolation schemes requires points which are outside the area of integration plus additional boundary lines.

As long as $|\alpha|$ and $|\beta|$ are <1 , only one passive boundary line is required, as is the case in an Eulerian integration. If values of $|\alpha|$ and $|\beta|$ are expected to be 2 or less, two passive boundary lines must be included, and so on. If space is at a premium, this represents a drawback of the semi-Lagrangian scheme as the size of the arrays has to be increased to include these extra values of the fields.

In this paper, the term "integration area" is used to refer to the fields at all of the points *exclusive* of the passive boundary lines.

If $(0, 0)$ represents the center of rotation, then, initially, the cone, which has a radius 4Δ ($\Delta = \Delta x = \Delta y$), and a height of 100, is centered at $(0, -8)$. The field ψ , which describes the cone, is updated at the 31×31 grid points ranging over $-15\Delta \leq x \leq 15\Delta$ and $-15\Delta \leq y \leq 15\Delta$. Passive boundary lines are added as needed.

The integration parameters are chosen as follows; mesh size $\Delta = 200$ km; angular velocity $\Omega = 0.3636 \times 10^{-4}$.

The velocities u and v are given by $u = -y\Omega$ and $v = x\Omega$ and the departure point (x_*, y_*) is chosen as

$$x_* = I\Delta x - u\Delta t, \quad y_* = J\Delta y - v\Delta t. \quad (35)$$

Obviously, this is not the correct departure point which is, in fact, at the point, expressed in polar coordinates,

$(r, \theta - \Omega\Delta t)$. Nevertheless, it is interesting to proceed in this fashion because, in a situation where the solution to the advection equation was not known, this would be the only way forward and would represent a first guess at the departure point.

With an integration timestep $\Delta t = 10$ min, the maximum value of α and β is 0.327. In practice this represents almost the maximum allowable timestep for a stable Arakawa-Adams-Bashforth integration. With this choice of timestep, the cone should have returned to its original position after 288 timesteps, and the chart of ψ should look like Fig. 3. The actual situation after this time is presented in Figs. 4a–d for the biquadratic, bicubic, biquartic and Arakawa-Adams-Bashforth schemes. (For the bilinear integration, the cone is reduced to a bump of height 9 units and hence the corresponding chart is not included.) Because $|\alpha|$ and $|\beta|$ are <1 , only one passive boundary line was needed for these integrations.

The biquartic interpolation gives results which are closest to the correct answer. The biquadratic scheme is marginally better than the Arakawa-Adams-Bashforth scheme and both have unpleasant "trailing waves." In the bicubic case, the height is no better represented than in the biquadratic case. However, the "trailing waves" are of smaller amplitude. These facts are in harmony with the discussion in Section 2 as can be seen by examining Figs. 1 and 2 in the region $|\alpha|$ and $|\beta|$ less than 0.327.

With a timestep of 1 h the values of α and β can range from -1.96 to $+1.96$ within the integration area and the cone should have returned to its original position in the plane after 48 timesteps. The actual situation after this time is presented in Figs. 5a, b, c for the biquadratic, bicubic and biquartic schemes. Again, and for the same reason, no bilinear chart is included. With this timestep, two passive boundary lines are needed. All of the semi-Lagrangian integrations are

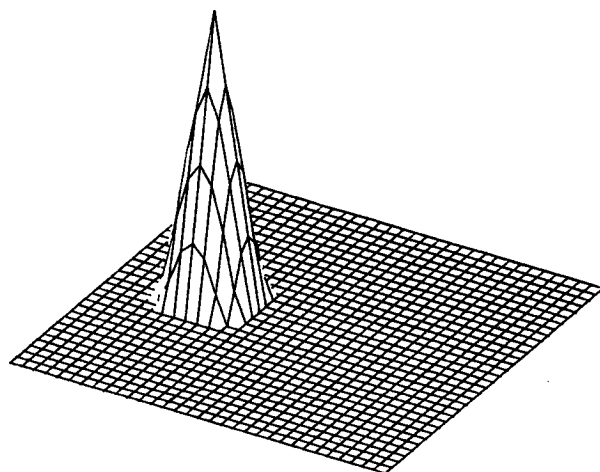


FIG. 3. Initial conditions.

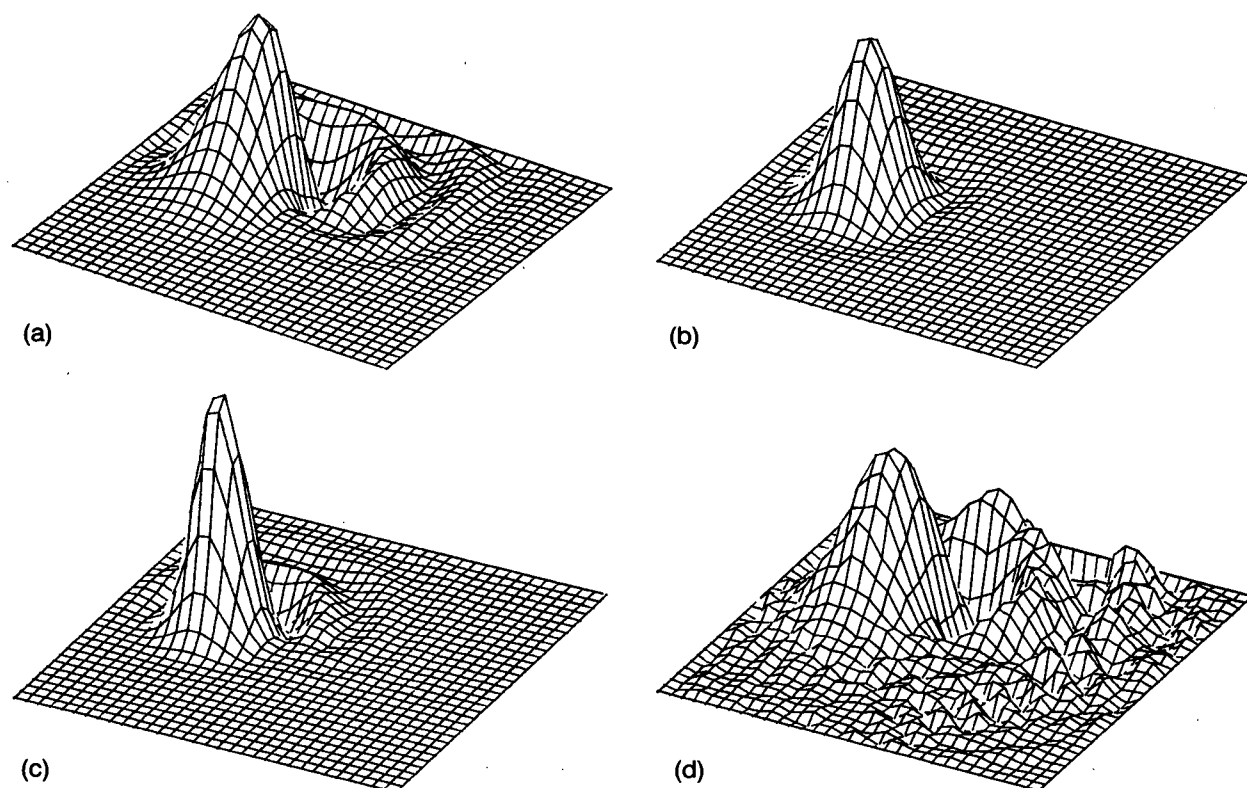


FIG. 4. Computed height after one revolution with a timestep of 10 min for (a) the biquadratic, (b) the bicubic, (c) the biquartic and (d) the Arakawa scheme.

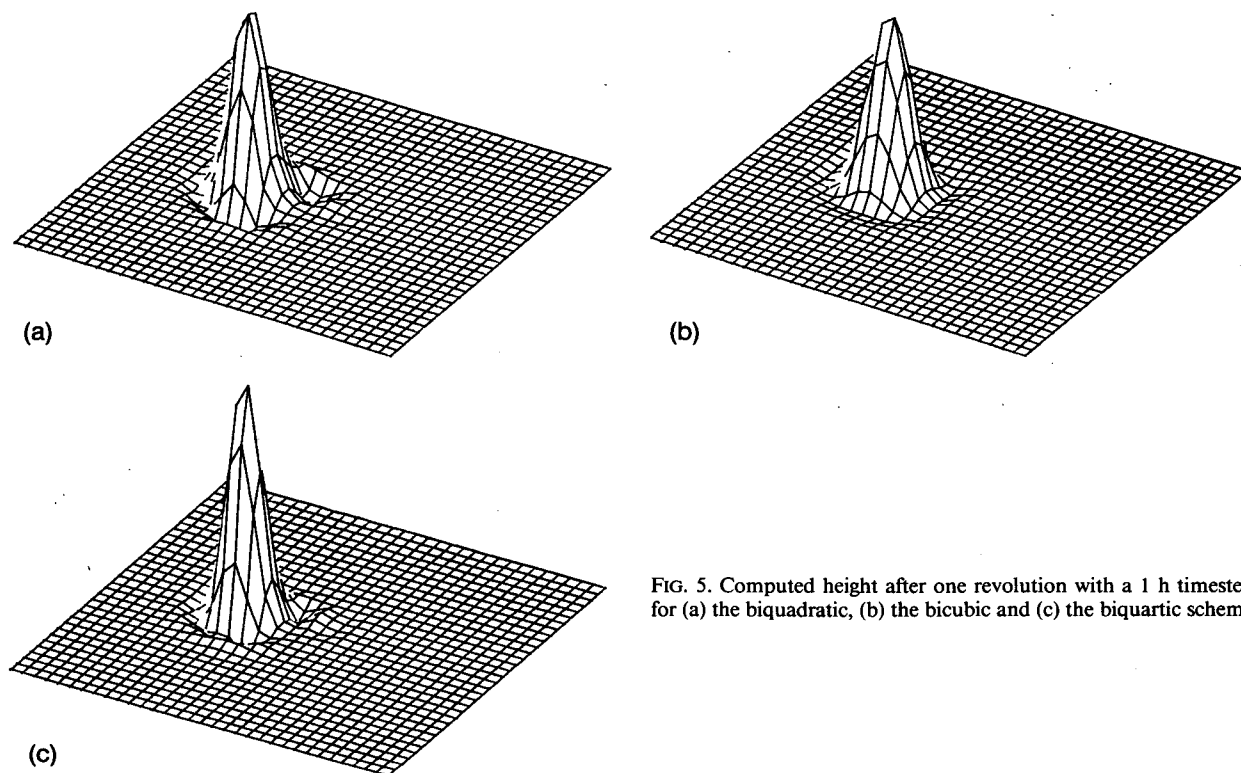


FIG. 5. Computed height after one revolution with a 1 h timestep, for (a) the biquadratic, (b) the bicubic and (c) the biquartic scheme.

stable, proving that the multiply-upstream idea works. Striking features of the biquadratic integration are the smallness of the trailing waves, and the fact that the height, shape and position of the cone are as good as, or better than the Eulerian integration with the 10 min timestep. Again, the biquartic integration is best, and the overall pattern seems in agreement with the analysis of Section 2.

Because the maximum timestep giving a stable integration for the Eulerian scheme was 10 min, the foregoing results, if they held up in a more realistic setting, would furnish the prospect of integrations, as good as, or better than the Eulerian one with a significant saving in computation time as can be seen from Table 1.

For all four interpolation schemes, stable and noise-free integrations have been run off with timesteps of up to 3 hours. This corresponds to a maximum value of α and β equal to 5.88. In principle, there is no reason why larger timesteps should not be used, provided the integration area is surrounded by the appropriate number of passive boundary lines. The main features of these integrations are presented in Tables 2a-e.

For timesteps up to 2 h, all of the schemes hold their own or show a modest improvement as far as the height and shape of the cone and the error in θ are concerned, and all except the bilinear integrations are better than the Arakawa-Adams-Bashforth scheme. However, the error in the radial position of the maximum increases with increasing timestep so that when $\Delta t = 2$ h it is already equal to 4.3Δ . Recall that the position of the maximum at $t = 0$ is 8Δ . The radial error is almost independent of which interpolation scheme is used, leading one to infer that it is caused by the incorrect choice of departure point.

The correct departure point (x_*, y_*) corresponding to the point $(r, \theta - \Omega\Delta t)$ is given by

$$\begin{aligned} x_* &= I\Delta x - \{u \sin\Omega\Delta t + v(1 - \cos\Omega\Delta t)\}/\Omega, \\ y_* &= J\Delta y - \{v \sin\Omega\Delta t - u(1 - \cos\Omega\Delta t)\}/\Omega. \end{aligned} \quad (36)$$

Proceeding as before, using the interpolation scheme given in Eq. (5) and the choice of interpolation points outlined in Eqs. (7) and (8) we can perform the integrations using the exact upstream points. The outcome of such integrations is presented in Tables 3a-d. Three main points emerge: First, for $\Delta t = 10$ min,

TABLE 1. Computation time per timestep of integration.

Scheme	CPU
Arakawa-Adams-Bashforth	0.109
Bilinear	0.089
Biquadratic	0.161
Bicubic	0.241
Biquartic	0.342

TABLE 2. Maxima, minima, radial error in units of Δ and angular error in degrees after one revolution in the plane using various timesteps, for the computational schemes.

Δt (hours)	Maximum	Minimum	Δr	$\Delta\theta$
<i>a. Bilinear</i>				
0.166	9	0	-1.7	-3°
1.0	18	0	-2.3	0°
2.0	23	0	-4.3	-3°
3.0	21	0	-5.3	-15°
<i>b. Biquadratic</i>				
0.166	55	-18	-0.7	-24°
1.0	64	-8	-3.0	+7°
2.0	64	-6	-4.7	0°
3.0	54	-4	-5.3	-22°
<i>c. Bicubic</i>				
0.166	55	-2	-0.7	-3°
1.0	60	-2	-2.7	0°
2.0	55	-1	-4.3	-5°
3.0	43	-2	-5.3	-22°
<i>d. Biquartic</i>				
0.166	80	-10	-0.7	-5°
1.0	80	-6	-2.7	0°
2.0	71	-5	-4.3	-5°
3.0	58	-5	-5.3	-22°
<i>e. Arakawa</i>				
0.166	53	-23	-0.3	-25°

using the exact departure point, Eq. (36), gives almost identical results to the approximate value, Eq. (35), for all four schemes; second, there is no radial error for all interpolation schemes for $\Delta t > 10$ mins; third, the maxima get higher for all interpolations as Δt is increased. Therefore, with the multiply-upstream semi-Lagrangian scheme the truncation error decreases with increasing timestep provided that the departure point (x_*, y_*) is accurately estimated.

When a set of advection equations whose solution is not known *a priori* is being integrated, it is customary to try to improve the estimate of the departure points by using an iterative scheme. For example, see Krishnamurti (1962), Mathur (1970), Robert (1981). Consider the following due to the last author. The $(k+1)$ th guess of the "halfway upstream" velocity is given in terms of the k th guess as follows:

$$\mathbf{v}_{IJ}^{k+1} = \mathbf{v}[I\Delta x - u_{IJ}^k\Delta t/2, J\Delta y - v_{IJ}^k\Delta t/2], \quad (37)$$

where

$$\mathbf{v}_{IJ}^0 = \mathbf{v}(I\Delta x, J\Delta y), \quad (38)$$

and the upstream point is given by

$$\mathbf{r}_* = -\Delta t \mathbf{v}_{IJ}^{k+1}. \quad (39)$$

TABLE 3. As in Table 2 but using the correct (exact) departure points, Eq. (36).

Δt (hours)	Maximum	Minimum	Δr	$\Delta \theta$
<i>a. Bilinear</i>				
0.166	9	0	-1.3	-2°
1.0	27	0	0.0	0°
2.0	41	0	0.0	0°
3.0	49	0	0.0	-3°
<i>b. Biquadratic</i>				
0.166	56	-18	-0.3	-22°
1.0	79	-7	0.0	0°
2.0	85	-2	0.0	0°
3.0	83	-4	0.0	0°
<i>c. Bicubic</i>				
0.166	56	-2	0.0	-2°
1.0	75	-1	0.0	0°
2.0	80	-1	0.0	0°
3.0	82	-1	0.0	0°
<i>d. Biquartic</i>				
0.166	81	-9	0.0	-5°
1.0	87	-3	0.0	0°
2.0	88	-2	0.0	0°
3.0	88	-3	0.0	0°

With $\Delta t = 7200$ s, the fifth iteration gives the same value of the velocity as the fourth iteration out to the fifth significant figure at $r = 8\Delta$. A series of integrations using the biquadratic scheme with one iteration was performed and the results are given in Table 4a. Comparing the results with those of the integrations using the exact upstream point (Table 3b), we can see that the maxima and minima differ by 1 unit of height and that the position of the apex differs by $\Delta r = 0.3$ and $\Delta \theta = 10^\circ$ for $\Delta t = 3$ h. Running with three iterations reduces these errors further while leaving the maxima and minima unchanged (Table 4b).

TABLE 4. As in Table 2, but using the iterative scheme, Eqs. (37)–(39) to estimate the departure point, for the biquadratic scheme.

Δt (hours)	Maximum	Minimum	Δr	$\Delta \theta$
<i>a. Biquadratic, one iteration</i>				
0.166	56	-18	0.0	-22°
1.0	80	-7	0.0	+3°
2.0	85	-3	0.0	+5°
3.0	83	-3	-3.0	+10°
<i>b. Biquadratic, three iterations</i>				
0.166	56	-18	0.0	-22°
1.0	80	-7	0.0	0°
2.0	84	-2	0.0	-3°
3.0	82	-5	0.0	-5°

TABLE 5. The calculated errors after one revolution.

	Δt			
	0.166	1.0	2.0	3.0
Δr_c	-0.5	-2.7	-4.4	-5.5
$\Delta \theta_c$	0°	-2°	-8°	-17°
$\Delta \theta_i$	0°	-1°	-2°	-5°

Let the apex be at (r_0, θ_0) at time $t = 0$. On the assumption of a perfect interpolation scheme, after n timesteps the Lagrangian scheme which uses the estimate of the upstream point given by Eq. (35) will advect the apex to a position (r_c, θ_c) given by

$$r_c = r_0(1 + \Omega^2 \Delta t^2)^{-n/2},$$

$$\theta_c = \theta_0 + n\tau, \quad (40)$$

where $\tau = \tan^{-1} \Omega \Delta t$. The values of $\Delta r_c = r_c - r_0$ and $\Delta \theta_c = n(\tau - \Omega \Delta t)$ are listed in Table 5 and comparing them with Δr and $\Delta \theta$ in Table 2, one sees that the error in the final position of the cone is mostly due to the error in estimating the upstream position, at least for $\Delta t \geq 1$ h. On the other hand, the iterative scheme outlined in Eqs. (37)–(39) converges to the solution $r_i = r_0$ and $\theta_i = \theta_0 + n\sigma$, where

$$\sigma = \tan^{-1} \{ \Omega \Delta t / (1 - \Omega^2 \Delta t^2 / 4) \}. \quad (41)$$

The values of $\Delta \theta_i = n(\sigma - \Omega \Delta t)$ are listed in Table 5 and again the agreement between this calculation of Δr_i and $\Delta \theta_i$ and the errors Δr and $\Delta \theta$ resulting from the integrations with three iterations (Table 4b), for $\Delta t \geq 1$ h, emphasizes the importance of accurately estimating the departure point when performing a Lagrangian integration.

4. Conclusions

The multiply-upstream, semi-Lagrangian scheme yields unconditionally stable integrations for velocity fields which are constant in time. To the extent that this result holds for nonconstant velocities it means that stability criteria need not be an overriding factor in choosing a timestep for integrating the advective equations. In fact, the larger the timestep used the more accurate the integration should be, provided the departure point can be accurately estimated. The fact that there is no exact method of finding this point for nonconstant velocities ultimately limits the size of the timestep which can be used.

The quadratic Lagrangian scheme was found to be as good as, or better than the Arakawa one for timesteps up to 1 h as far as position, shape and height of the cone are concerned, using the noniterative scheme. To maintain or improve on this standard of accuracy with larger timesteps, a more accurate estimate of the departure point was required and the iterative scheme fulfilled this function.

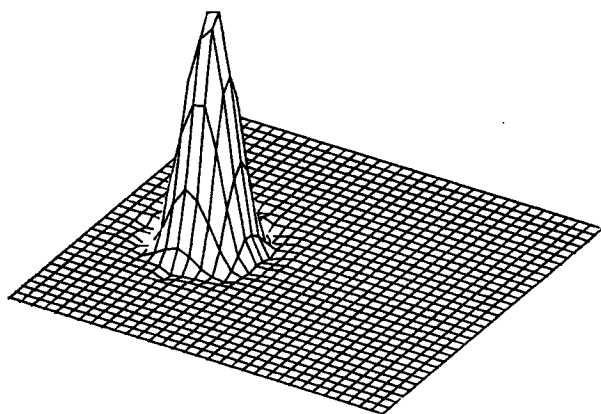


FIG. 6. Computed height after one revolution with a 3 h timestep for the biquadratic scheme with one iteration.

In the example shown, the biquadratic iterative scheme with one iteration and a timestep of 3 h gave a more accurate integration than the Arakawa scheme with a timestep of 10 min (see Fig. 6).

In a physically more authentic situation u and v as well as ψ would be advected at each timestep. This would entail 5 interpolations per 3 h timestep for the Lagrangian scheme versus an Arakawa-type integration for each of u , v , and ψ per 10 min timestep, making the former approximately five times as efficient as the latter.

In one respect the Arakawa scheme is superior to the Lagrangian schemes in that the following quantities are conserved in time,

$$\sum_{i,j} \{\psi_{ij}^n\}^2 \quad \text{and} \quad \sum_{i,j} \psi_{ij}^n,$$

the former in the limit as $\Delta t \rightarrow 0$, Δx held fixed and the latter for all Δt and Δx . For the Lagrangian schemes, only the latter property holds and then only for constant advecting velocity.

With regard to the relative merits of the interpolation schemes, the bilinear yields excessively smoothed predictions; the biquadratic gives results which are as good as, but less expensive than the Arakawa-Adams-Bashforth integration; and the bicubic and biquartic seem expensive for the small increase in accuracy afforded, at least for large timesteps, in the idealized calculations examined.

APPENDIX

The D and W Equations

$$D_1\psi = \partial\psi/\partial t + u\partial\psi/\partial x,$$

$$D_2\psi = D_1\psi + (\partial^2\psi/\partial t^2 - u^2\partial^2\psi/\partial x^2)\Delta t/2,$$

$$D_3\psi = D_2\psi + (\partial^3\psi/\partial t^3 + u^3\partial^3\psi/\partial x^3)\Delta t^2/6,$$

$$D_4\psi = D_3\psi + (\partial^4\psi/\partial t^4 - u^4\partial^4\psi/\partial x^4)\Delta t^3/24.$$

Because $d\psi/dt = 0$, it follows that $D_r\psi = 0$ for all r .

$$W_1'(\hat{\alpha}, p) = (1 - \hat{\alpha})(-p)^j + \hat{\alpha}(-p - 1)^j,$$

$$W_2'(\hat{\alpha}, p) = \hat{\alpha}(1 + \hat{\alpha})(-p - 1)^j/2 \\ + (1 - \hat{\alpha})(1 + \hat{\alpha})(-p)^j - \hat{\alpha}(1 - \hat{\alpha})(-p + 1)^j/2,$$

$$W_3'(\hat{\alpha}, p) = -\hat{\alpha}(1 - \hat{\alpha}^2)(-p - 2)^j/6 \\ + \hat{\alpha}(1 + \hat{\alpha})(2 - \hat{\alpha})(-p - 1)^j/2 \\ + (1 - \hat{\alpha}^2)(2 - \hat{\alpha})(-p)^j/2 \\ - \hat{\alpha}(1 - \hat{\alpha})(2 - \hat{\alpha})(-p + 1)^j/6,$$

$$W_4'(\hat{\alpha}, p) = -\hat{\alpha}(1 - \hat{\alpha}^2)(2 + \hat{\alpha})(-p - 2)^j/24 \\ + \hat{\alpha}(2 - \hat{\alpha}^2)(1 + \hat{\alpha})(-p - 1)^j/6 \\ + \hat{\alpha}(1 - \hat{\alpha}^2)(2 - \hat{\alpha})(-p + 2)^j/24 - \hat{\alpha}(2 - \hat{\alpha}^2) \\ \times (1 - \hat{\alpha})(-p + 1)^j/6 + (2 - \hat{\alpha}^2)(1 - \hat{\alpha}^2)(-p)^j/4.$$

REFERENCES

- Bates, J. R., and A. McDonald, 1982: Multiply-upstream, semi-Lagrangian advective schemes: Analysis and application to a multi-level primitive equation model. *Mon. Wea. Rev.*, **110**, 1831-1842.
- Carnahan, B., H. A. Luther and J. O. Wilkes, 1969: *Applied Numerical Methods*. Wiley, 604 pp.
- Crowley, W. P., 1969: Numerical advection experiments. *Mon. Wea. Rev.*, **96**, 1-11.
- Krishnamurti, T. N., 1962: Numerical integrations of primitive equations by a quasi-Lagrangian advective scheme. *J. Appl. Meteor.*, **1**, 508-521.
- Mathur, M. B., 1970: A note on an improved quasi-Lagrangian advective scheme for the primitive equations. *Mon. Wea. Rev.*, **98**, 214-219.
- Molenkamp, C. R., 1968: Accuracy of finite difference methods applied to the advection equation. *J. Appl. Meteor.*, **7**, 160-167.
- Orszag, S. A., 1971: Numerical simulation of incompressible flows within simple boundaries: Accuracy. *J. Fluid Mech.*, **49**, 75-112.
- Richtmeyer, R. D., and K. W. Morton, 1967: *Difference Methods for Initial Value Problems*, 2nd ed., Wiley, p. 45.
- Robert, A., 1981: A stable numerical integration scheme for the primitive meteorological equations. *Atmos. Ocean*, **19**, 35-46.
- , 1982: Semi-Lagrangian and semi-implicit numerical integration scheme for the primitive meteorological equations. *J. Meteor. Soc. Japan*, **60**, 319-324.